

ASSIGNMENT - 2 - AI

Q1) Unsupervised Learning by hand

1.1) PCA on four points

$$A = (1, 3) \quad B = (3, 1) \quad C = (5, 3) \quad D = (7, 5)$$

Sample	x_1	x_2
A	1	3
B	3	1
C	5	3
D	7	5

a) $n = 4$ and $p = 2$
(samples) (features)

$$X = \begin{bmatrix} 1 & 3 \\ 3 & 1 \\ 5 & 3 \\ 7 & 5 \end{bmatrix}$$

Step 1: mean of x_1

$$\bar{x}_1 = \frac{1 + 3 + 5 + 7}{4} = \frac{16}{4} = \boxed{4}$$

Step 2: mean of x_2

$$\bar{x}_2 = \frac{3 + 1 + 3 + 5}{4} = \frac{12}{4} = \boxed{3}$$

$\therefore$ mean
(4, 3)

$\therefore$ The centered points will be

$$A - \mu = (1 - 4, 3 - 3) = (-3, 0)$$

$$B - \mu = (3 - 4, 1 - 3) = (-1, -2)$$

$$C - \mu = (5 - 4, 3 - 3) = (1, 0)$$

$$D - \mu = (7 - 4, 5 - 3) = (3, 2)$$

$$\therefore \text{Centered dataset } (X_c) = \begin{bmatrix} -3 & 0 \\ -1 & -2 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}$$

$$\therefore X_c^T = \begin{bmatrix} -3 & -1 & 1 & 3 \\ 0 & -2 & 0 & 2 \end{bmatrix}$$

b) Covariance Matrix (finding by $n=4$)

$$\therefore \text{The population covariance matrix} = \frac{1}{n} X_c^T X_c$$

$$\therefore = \frac{1}{4} \begin{bmatrix} -3 & -1 & 1 & 3 \\ 0 & -2 & 0 & 2 \end{bmatrix}_{2 \times 4} \begin{bmatrix} -3 & 0 \\ -1 & -2 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}_{4 \times 2}$$

$$= \frac{1}{4} \begin{bmatrix} 9+1+1+9 & 0+2+0+6 \\ 0+2+0+6 & 0+4+0+4 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 20 & 8 \\ 8 & 8 \end{bmatrix}$$

$$(\Sigma) \text{ Covariance Matrix} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}_{2 \times 2}$$

$\therefore$ Variance of $x_1 = 5$
 " " $x_2 = 2$
 Covariance = 2

c) Compute Eigenvalues and Eigen vectors.

$$\therefore \det(\Sigma - \lambda I) = 0$$

$$\therefore \Sigma - \lambda I = \begin{bmatrix} 5-\lambda & 2 \\ 2 & 2-\lambda \end{bmatrix}$$

$\therefore$ We need to prove

$$\det \begin{bmatrix} 5-\lambda & 2 \\ 2 & 2-\lambda \end{bmatrix} = 0$$

$$\lambda = \frac{1}{2} \left(\begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix} \pm \sqrt{(5-2)^2 + 4(2)^2} \right) = \frac{1}{2} [10-4] \pm \sqrt{9+16}$$

$$= \frac{1}{2} [6] \pm \sqrt{25}$$

$$\neq 5$$

$$\therefore (5-\lambda)(2-\lambda) - (2)(2) = 0$$

$$10 - 5\lambda - 2\lambda + \lambda^2 - 4 = 0$$

$$\therefore \lambda^2 - 7\lambda + 6 = 0$$

$$(\lambda-6)(\lambda-1) = 0$$

$$\therefore \boxed{\lambda=6} \text{ \& \& } \boxed{\lambda=1}$$

$$\begin{array}{c} -7 \\ \swarrow \searrow \\ -6 \quad -1 \end{array}$$

Hence the two eigen vectors are: $\boxed{\lambda_1=6}$ $\boxed{\lambda_2=1}$

Now, calculating eigen values for the corresponding vectors.

Eigen Vector for $\lambda_1=6$

$$(\Sigma - 6I) v_1 = 0$$

$$\therefore \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\therefore -v_1 + 2v_2 + 2v_1 - 4v_2 = 0$$

$$v_1 - 2v_2 = 0$$

$$\therefore v_1 = 2v_2$$

$$\text{Let, } \boxed{v_2=1}, \therefore \boxed{v_1=2}$$

$$\therefore v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$\therefore$ Normalize, To get unit eigenvector.

$$\therefore v_1 = \frac{1}{\sqrt{5}} (2, 1)$$

$$\lambda_1 = 6$$

Eigen Vector for $\lambda_2=1$

$$(\Sigma - 1I) v_2 = 0$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$4v_1 + 2v_2 + 2v_1 + v_2 = 0$$

$$6v_1 + 3v_2 = 0$$

$$\therefore \cancel{3}v_2 = \cancel{-2}v_1$$

$$v_2 = -2v_1$$

$$\text{Let, } \boxed{v_1=1}, \therefore \boxed{v_2=-2}$$

$$\therefore v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$v_2 = \frac{1}{\sqrt{5}} (1, -2)$$

$$\lambda_2 = 1$$

d) Explained variance ratio.

$$EVR_i = \frac{\lambda_i}{\sum_{j=1}^k \lambda_j}$$

Our eigenvalues are 6 and 1.

$$\therefore \text{Total variance} : \underline{6+1=7}$$

PC1

$$EVR_1 = \frac{6}{7}$$

$$\therefore \boxed{EVR_1 = 85.714\%}$$

PC2

$$EVR_2 = \frac{1}{7}$$

$$\boxed{EVR_2 = 14.286\%}$$

$\therefore$ PC1 is the direction that captures the major amount of variance.

e) Project D onto PC1

$$\begin{aligned} \frac{D(7,1)}{D(3,2)} &\rightarrow \text{PC1} \\ \rightarrow D(3,2) &\rightarrow \text{PC1} \end{aligned}$$

$$\text{PC}_1 \text{ unit vector, } \boxed{v_1 = \frac{1}{\sqrt{5}} (2, 1)}$$

$$\begin{aligned} z_D &= D_c \cdot v_1 \\ &= (3, 2) \cdot \frac{1}{\sqrt{5}} (2, 1) \end{aligned}$$

$$= \frac{3(2) + 2(1)}{\sqrt{5}} = \frac{8}{\sqrt{5}} \approx 3.578 //$$

f) Now make x_2 ten times larger.

Originally, $A=(1,3)$; $B=(3,1)$; $C=(5,3)$; $D=(7,5)$

$$A=(1,30) \quad B=(3,10) \quad C=(5,30) \quad \& \quad D=(7,50)$$

new means $\bar{x}_1 = 4$ $\bar{x}_2 = 30$

$$A_c = (-3, 0); B_c = (-1, -20); C_c = (1, 0); D_c = (3, 20)$$

new new new new

$$\therefore \text{Covariance Matrix: } \Sigma_{\text{scaled}} = \begin{bmatrix} 5 & 20 \\ 20 & 200 \end{bmatrix}$$

$\therefore$ Similarly using previous steps

$$\underline{\text{EVR}_1 = 98.55\%}$$

PC1 points primarily along $\boxed{x_2}$

Correlation-matrix eigen values of original data.

$$1 + \sqrt{0.4} = \underline{1.6325} \quad \& \quad 1 - \sqrt{0.4} = \underline{0.3675}$$

1.2) One run of Lloyd's Algorithm:

Point x y

A 0 0

B 2 0

C 3 1

D 8 0

E 9 1

F 10 0

Kmean: - K = 2

and initial centroids

$c_1 = A = (0, 0)$ & $c_2 = B = (2, 0)$

a) Iteration 1

For k-means, euclidean distance.

$$d^2((x, y), c) = (x - c_x)^2 + (y - c_y)^2$$

Our ~~initial~~ initial centroids are:

$c_1 = (0, 0)$ & $c_2 = (2, 0)$

→ Squared Distances

Point	d^2 to $c_1 = (0, 0)$	d^2 to $c_2 = (2, 0)$	
A(0, 0)	$(0)^2 + (0)^2 = 0$	$(2)^2 + (0)^2 = 4$	→ c_1
B(2, 0)	$(2)^2 + (0)^2 = 4$	$(0)^2 + (0)^2 = 0$	→ c_2
C(3, 1)	$3^2 + 1^2 = 10$	$1^2 + 1^2 = 2$	→ c_2
D(8, 0)	$8^2 + 0^2 = 64$	$6^2 + 0^2 = 36$	→ c_2
E(9, 1)	$9^2 + 1^2 = 82$	$7^2 + 1^2 = 50$	→ c_2
F(10, 0)	$10^2 + 0^2 = 100$	$8^2 + 0^2 = 64$	→ c_2

$c_1 \rightarrow \{A\}$

$c_2 \rightarrow \{B, C, D, E, F\}$

Update The centroids

for cluster 1, we only have A:

$$C_1 = \frac{(0,0)}{1} \longrightarrow \boxed{C_1 = (0,0)}$$

$$B = (2,0), C = (3,1), D = (8,0), E = (9,1), F = (10,0)$$

Calculate x-coordinate

$$\frac{2+3+8+9+10}{5} = \frac{32}{5} = 6.4$$

Calculate y-coordinate

$$\frac{0+1+0+1+0}{5} = \frac{2}{5} = 0.4$$

$$\therefore \boxed{C_2 = (6.4, 0.4)}$$

$$\therefore \underline{C_1 = (0,0)} \quad \& \quad \underline{C_2 = (6.4, 0.4)}$$

b) Iteration 2

	$C_1 = (0,0)$	$C_2 = (6.4, 0.4)$	
Point A	$d^2(A, C_1) = 0$	$d^2(B, C_1) = 2^2 = 4$	Point B
	$d^2(A, C_2) = (0-6.4)^2 + (0-0.4)^2$	$d^2(B, C_2) = (2-6.4)^2 + (0-0.4)^2$	
	$d^2(A, C_2) = 40.96 + 0.16 = 41.12$	$= 19.36 + 0.16 = 19.52$	
So A remains on C_1		So B changes from C_2 to C_1	

Point D

$$d^2(D, c_1) = 8^2 = \underline{64}$$

$$d^2(D, c_2) = (8-6.4)^2 + (0-0.4)^2 \\ = 2.56 + 0.16$$

$$\boxed{d^2(D, c_2) = 2.72}$$

Point C

$$d^2(C, c_1) = 3^2 + 1^2 = \underline{10}$$

$$d^2(C, c_2) = (3-6.4)^2 + (1-0.4)^2 \\ = 11.56 + 0.36$$

$$\boxed{d^2(C, c_2) = 11.92}$$

~~Point E~~

Point E

$$d^2(E, c_1) = 9^2 + 1^2 = \underline{82}$$

$$d^2(E, c_2) = (9-6.4)^2 + (1-0.4)^2 \\ = 6.76 + 0.36$$

$$\boxed{d^2(E, c_2) = 7.12}$$

Point F

$$d^2(F, c_1) = 10^2 = 100$$

$$d^2(F, c_2) = (10-6.4)^2 + (0-0.4)^2 \\ = 12.96 + 0.16$$

$$\boxed{d^2(F, c_2) = 13.12}$$

Iteration Table 2

Point	d^2 to c_1	d^2 to c_2	
A	0	41.12	$\rightarrow c_1$
B	4	19.52	$\rightarrow c_1$
C	10	11.92	$\rightarrow c_1$
D	64	2.72	$\rightarrow c_2$
E	82	7.12	$\rightarrow c_2$
F	100	13.12	$\rightarrow c_2$

$$\therefore C_1 = \{A, B, C\}$$

$$\& C_2 = \{D, E, F\}$$

c) Running the assignment step once more.

Cluster 1:

$$A = (0, 0) \quad B = (2, 0) \quad C = (3, 1)$$

$$C_1 = \left(\frac{0 + 2 + 3}{3}, \frac{0 + 0 + 1}{3} \right)$$

$$C_1 = \left(\frac{5}{3}, \frac{1}{3} \right)$$

Cluster 2:

$$D = (8, 0); \quad E = (9, 1), \quad F = (10, 10)$$

$$C_2 = \left(\frac{8 + 9 + 10}{3}, \frac{0 + 1 + 0}{3} \right)$$

$$C_2 = \left(9, \frac{1}{3} \right)$$

$$A, B, C \rightarrow C_1$$

$$D, E, F \rightarrow C_2$$

There is no change in labels.

$\therefore$ The algorithm has converged.

d) Silhouette of A

$$a(A) = \frac{d(A,B) + d(A,C)}{2} = \frac{2 + \sqrt{10}}{2} \approx \underline{2.58}$$

$$b(A) = \frac{d(A,D) + d(A,E) + d(A,F)}{3} = \frac{8 + \sqrt{82} + 10}{3} = \underline{9.02}$$

$$s(A) = \frac{b - a}{\max(a, b)} = \underline{\underline{0.71}}$$

e) Kmeans + probabilities

$$B: 4, C: 10, D: 64, E: 82, F: 100$$

$$4 + 10 + 64 + 82 + 100 = \underline{\underline{260}}$$

$$\therefore P(B) = \frac{4}{260} = 1.54\% \quad P(C) = 3.85\% \quad P(D) = 24.62\%$$

$$P(E) = 31.54\% \quad \& \quad P(F) = 38.64\%$$

$$\therefore P(D, E, F) = \underline{\underline{94.62\%}}$$

1.3) Single Vs Complete Linkage

a) Distance Matrix

	A	B	C	D	E
A	0	4	6	7	14
B	4	0	2	3	10
C	6	2	0	1	8
D	7	3	1	0	7
E	14	10	8	7	0

b) Single linkage

$$\begin{array}{ccccccc}
 C-D & \rightarrow & B-(CD) & \rightarrow & A-(BCD) & \rightarrow & E-(ABCD) \\
 (1) & & (2) & & (4) & & (7)
 \end{array}$$

$$\therefore \text{heights} \rightarrow 1, 2, 4, 7$$

c) Complete linkage

$$\begin{array}{ccccccc}
 C-D & \rightarrow & B-(CD) & \rightarrow & A-(BCD) & \rightarrow & E-(ABCD) \\
 (1) & & (3) & & (7) & & (14)
 \end{array}$$

$$\text{heights} \rightarrow 1, 3, 7, 14$$

d) Single linkage

$$\begin{array}{lcl}
 K=2 & 4 < h < 7 \\
 K=3 & 2 < h < 4
 \end{array}$$

Complete linkage

$$\begin{array}{lcl}
 K=2: & 7 < h < 14 \\
 K=3: & 3 < h < 7
 \end{array}$$

$\therefore$ for $K=2$
 $\{A, B, C, D\}; \{E\}$

for $K=3$
 $\{A\}; \{B, C, D\}; \{E\}$

Linkage changes the merge heights but not the $K=2$ or $K=3$ partition for this dataset

e) Single Linkage because it connect nearby points along the curved chains of the crescents, whereas complete linkage favors more compact clusters.

1.4) Two conceptual questions
a)